

TEAM PROJECT PITCH AND ROLES

Final Project Direction, Team Commitment, and Execution Plan

Purpose. Define one feasible AI or machine-learning project and show how your team will complete it. Your pitch must make the goals, data, methodology, timeline, evaluation, and each member's contribution clear.

Team lock. This submission confirms your final team name and membership. Teams must have 4-6 members. You may belong to only one team. Team changes will not be allowed after this submission, so choose carefully. If your name appears on more than one team, you will receive a zero. Check the course team-groups page before submitting.

1 Final Team Confirmation

TEAM NAME	FINAL TEAM MEMBERS Full names as listed in Canvas	SUBMISSION DATE
Insert Team Name Here	1. Dakarai Lanes 2. Rickey Johnson 3. John Sampson 4. Amaka Okoro 5. Premajyoti Gammon	September 27, 2026

Roster confirmation: By submitting, the team confirms that every person listed agreed to join this team, appears on no other team, and is included in the Canvas team-groups page.

2 Project Pitch

FIELD	TEAM RESPONSE
Working title	Smart Recycling Coach
Problem and user	Students and campus facilities staff need quick, consistent disposal guidance for common items. The project helps a user decide whether an item belongs in recycling, trash, or special handling.
Project objective	Train and test a three-class image classifier that gives a disposal suggestion from a photo. The team will compare a fine-tuned model with a majority-class baseline on held-out images and demonstrate the result in a small web app.
AI or ML role	Fine-tune a MobileNetV3 image classifier pretrained on ImageNet for recycling, trash, and special-handling labels. Image classification is appropriate because material and item appearance vary too much for a fixed keyword or rule system.
Final deliverables	A documented image dataset, label guide, training notebook and saved model, baseline and model results with a confusion matrix, Streamlit demo, and presentation materials.

3 Specific Aims or Goals

AIM	SPECIFIC GOAL	MEASURABLE RESULT OR OUTPUT	PRIMARY OWNER(S)
1	Create a balanced three-class image dataset and label guide.	At least 900 usable images, class counts, source metadata, and a reviewed labeling guide.	John Sampson, Amaka Okoro, Premajyoti Gammon
2	Build a baseline and fine-tune a MobileNetV3 classifier.	Reproducible training notebook, saved checkpoint, and held-out predictions for both comparisons.	Dakarai Lanes, Rickey Johnson
3	Evaluate errors and integrate the model into a usable demo.	Accuracy, macro F1, per-class recall, confusion matrix, and a demo tested on at least 10 new images.	Dakarai Lanes, Rickey Johnson, Amaka Okoro, Premajyoti Gammon

4 Data Plan

Be specific. Naming only a website, repository, or general data type is not enough.

DATA QUESTION	TEAM RESPONSE
Dataset or source	TACO: Trash Annotations in Context (https://github.com/pedropro/TACO); team-taken photos of common clean items; EPA recycling guidance (https://www.epa.gov/recycle/how-do-i-recycle-common-recyclables).
Access or creation	John will download TACO and retain images fitting three broad labels. Each member will contribute about 25 photos of clean items. The team will label with one guide and Premajyoti will coordinate a second review of disagreements. No faces, people, personal records, or restricted locations will be collected; TACO attribution and license terms will be retained.
Expected structure	Target 900-1,200 RGB JPG images. Unit: one photo of one item. Labels: recycling, trash, or special handling. Inputs are resized images; metadata records source, label, and reviewer.
Preparation and splits	Remove duplicates, unreadable images, and inconsistent labels; resize images; and review a sample of each class. Keep near-duplicate items in the same split, then use stratified 70/15/15 train/validation/test splits. Apply augmentation only to training images.
Limits and ethics	Recycling rules vary by location, so the demo gives general guidance and does not replace campus policy. The team will avoid identifiable images, respect data licenses, avoid hazardous-waste decisions, and report errors by class and lighting condition.

5 Methodology and Evaluation

METHOD	TEAM RESPONSE
Workflow	Load and document images with Python, pandas, Pillow, and PyTorch. Establish a majority-class baseline, then fine-tune MobileNetV3 with the same train/validation split. Export predictions and model confidence, then connect the saved model to a simple Streamlit upload interface.
Evaluation	Use held-out accuracy, macro F1, per-class recall, and a confusion matrix. Compare MobileNetV3 with the majority-class baseline. Success means macro F1 of at least 0.70, improvement over baseline, and correct response flow for at least 9 of 10 demo images selected after training.
Feasibility and risk	The greatest risk is too few or inconsistent images for a difficult class. Fallback: use a clean TACO-only three-class subset and deliver the dataset card, baseline, trained model or error analysis, and a static prediction demo.

6 Timeline and Milestones

TARGET DATE	AIM OR PHASE	CONCRETE MILESTONE OR OUTPUT	OWNER(S)
Oct. 4	Data acquisition or creation	Source log, draft label guide, and first 500 screened images.	John, Amaka, Premajyoti
Oct. 11	Data preparation and baseline	Clean dataset, 70/15/15 split manifest, and majority-class baseline result.	Rickey, John
Oct. 25	Modeling or system build	MobileNetV3 training notebook, checkpoint, and validation metrics.	Dakarai, Rickey
Nov. 8	Evaluation and revision	Held-out report, confusion matrix, error review, and revised labels or settings.	Dakarai, Amaka, Premajyoti
Nov. 22	Final integration and presentation	Streamlit demo, final repository, presentation, and contribution summary.	All members

7 Individual Roles and Contributions

TEAM MEMBER	PRIMARY ROLE	SPECIFIC CONTRIBUTION OR DELIVERABLE	DEPENDENCY OR PARTNER
Dakarai Lanes	Project lead and model developer	Own the training pipeline, baseline comparison, model checkpoint, and metrics summary.	Rickey for split files and validation data; Amaka for demo integration.
Rickey Johnson	Data engineer and validation analyst	Clean images, maintain split manifests, check duplicates, and produce prediction and error tables.	John for source and label records; Dakarai for model outputs.
John Sampson	Data acquisition and labeling lead	Download and screen source data, coordinate team photos, maintain label guide and data documentation.	Amaka for label review; Rickey for quality checks.
Amaka Okoro	Demo and evaluation lead	Build the Streamlit interface, test user flow, review labels, and assemble presentation visuals.	Dakarai for saved model; John for label guidance.
Premajyoti Gammon	Quality review and documentation	Conduct a second label review, maintain the data card and source log, and summarize error patterns for the final report.	John for initial labels; Rickey and Dakarai for prediction results.

Submission checklist. One PDF per team; maximum four pages; all names match Canvas; no member appears on another team; every aim has a measurable output; all data sources and creation steps are named; methodology and evaluation are clear; every milestone has a date and owner; every member has a specific contribution.